PYTHON TEAM PROJECT

TOPIC: HOUSE PREDICTION

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****FLUME****

Cell 1: Create the Flume Configuration

In [10]:
```
%%writefile flume_hdfs_csv.conf

# Name the components
a1.sources = r1
a1.sinks = k1
a1.channels = c1

# SOURCE: Listen to Python on port 44445
a1.sources.r1.type = netcat
a1.sources.r1.bind = localhost
a1.sources.r1.port = 44445

# SINK: HDFS (Your new folder)
a1.sinks.k1.type = hdfs
a1.sinks.k1.hdfs.path = hdfs://localhost:9000/user/praveen/PYTHON_BIGDATA.ipynb/
a1.sinks.k1.hdfs.fileType = DataStream

# Roll Settings (Control when to save the file)
# We want to see the file quickly, so roll after 100 events or 30 seconds
a1.sinks.k1.hdfs.rollCount = 100
a1.sinks.k1.hdfs.rollInterval = 30
a1.sinks.k1.hdfs.rollSize = 0

# CHANNEL: Memory
a1.channels.c1.type = memory
```

```
a1.channels.c1.capacity = 1000
a1.channels.c1.transactionCapacity = 100

# Bind them
a1.sources.r1.channels = c1
a1.sinks.k1.channel = c1
```

```
Writing flume_hdfs_csv.conf
```

Cell 3: Python Streamer (Sends 100 CSV Rows)

In [11]:

```
import socket
import time
import random

# Configuration
HOST = 'localhost'
PORT = 44445
DATA_LIMIT = 100  # Exactly 100 rows

def generate_boston_csv_row():
    """Generates a random row matching the Boston Dataset columns"""
    # Columns: CRIM, ZN, INDUS, CHAS, NOX, RM, AGE, DIS, RAD, TAX, PTRATIO, B, LSTAT, MEDV
    crim = round(random.uniform(0.006, 88.9), 5)
    zn = round(random.uniform(0, 100), 1)
    indus = round(random.uniform(0.4, 27.7), 2)
    chas = random.choice([0, 1])
    nox = round(random.uniform(0.38, 0.87), 3)
    rm = round(random.uniform(3.5, 8.7), 2)
    age = round(random.uniform(2.9, 100), 1)
    dis = round(random.uniform(1.1, 12.1), 2)
    rad = random.randint(1, 24)
    tax = random.randint(187, 711)
    ptratio = round(random.uniform(12.6, 22.0), 1)
    b = round(random.uniform(0.32, 396.9), 2)
    lstat = round(random.uniform(1.73, 37.9), 2)
    medv = round(random.uniform(5.0, 50.0), 1) # This is the Price

    # Create CSV string
    return f"{crim},{zn},{indus},{chas},{nox},{rm},{age},{dis},{rad},{tax},{ptratio},{b},{lstat},{medv}"
```

```python
print(f"--- Connecting to Flume on {HOST}:{PORT} ---")

try:
    s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    s.connect((HOST, PORT))
    print(f"Connected! Sending {DATA_LIMIT} CSV rows...")

    for i in range(DATA_LIMIT):
        # 1. Generate Data
        csv_row = generate_boston_csv_row()

        # 2. Add Newline (Important for Flume to know it's a new row)
        message = csv_row + "\n"

        # 3. Send
        s.sendall(message.encode('utf-8'))

        # Fast sleep (0.05s) so it finishes in about 5 seconds
        time.sleep(0.05)

    print(f"--- Done! Sent exactly {DATA_LIMIT} rows. ---")
    s.close()

except ConnectionRefusedError:
    print("❌ ERROR: Could not connect. Is the Flume agent running?")
except Exception as e:
    print(f"❌ ERROR: {e}")
```

```
--- Connecting to Flume on localhost:44445 ---
Connected! Sending 100 CSV rows...
--- Done! Sent exactly 100 rows. ---
```

Cell 4: Verify HDFS Content

In [13]:

```python
import os

# Define the HDFS command
hdfs_path = "/user/praveen/PYTHON_BIGDATA.ipynb/"
print(f"Checking HDFS path: {hdfs_path}")

# List files
```

```
list_cmd = f"hdfs dfs -ls {hdfs_path}"
print(f"Running: {list_cmd}\n")
os.system(list_cmd)

print("\n--- Content Preview (Head of one file) ---")
# This command reads the first few lines of the data to prove it is CSV
cat_cmd = f"hdfs dfs -cat {hdfs_path}* | head -n 5"
os.system(cat_cmd)
```

```
Checking HDFS path: /user/praveen/python_flume_try/
Running: hdfs dfs -ls /user/praveen/python_flume_try/

Found 1 items
-rw-r--r--   1 praveen supergroup       7098 2025-11-29 10:35 /user/praveen/python_flume_try/FlumeData.1764412497070

--- Content Preview (Head of one file) ---
11.88608,69.1,9.31,1,0.629,7.7,55.4,3.3,8,416,21.8,72.51,8.33,19.3
13.70053,18.7,3.91,1,0.564,7.14,53.5,11.08,5,327,12.8,129.24,12.3,36.6
57.8441,98.1,22.27,0,0.4,7.0,98.3,8.2,22,647,13.4,65.68,26.97,37.0
6.46763,54.8,24.9,1,0.812,4.76,88.9,10.68,21,698,20.2,358.65,29.28,48.8
39.50462,84.5,5.38,1,0.722,6.21,49.5,7.73,22,490,17.6,320.15,33.69,34.3
cat: Unable to write to output stream.
```

Out[13]: 0

ML USING PYSpark

Cell 1: Imports & Data Loading

In [28]:
```
import os
import sys
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from pyspark.sql import SparkSession
from pyspark.sql.functions import col
from pyspark.ml.feature import VectorAssembler, MinMaxScaler
from pyspark.ml.regression import RandomForestRegressor
from pyspark.ml.evaluation import RegressionEvaluator
from pyspark.ml import Pipeline
```

```python
# Initialize Spark
print("--- Initializing Spark Session ---")
spark = SparkSession.builder \
    .appName("Boston_High_Accuracy") \
    .master("local[*]") \
    .getOrCreate()
print(f"✅ Spark Version: {spark.version}")

# Load Data
hdfs_path = "hdfs://localhost:9000/user/praveen/bostan_dataset/boston.csv"
print(f"\n--- Loading Data from {hdfs_path} ---")
df = spark.read.csv(hdfs_path, header=True, inferSchema=True)

row_count = df.count()
print(f"✅ Data Loaded Successfully. Total Rows: {row_count}")
print("Schema:")
df.printSchema()
```

```
--- Initializing Spark Session ---
✅ Spark Version: 4.0.1

--- Loading Data from hdfs://localhost:9000/user/praveen/bostan_dataset/boston.csv ---
25/11/29 11:22:12 WARN SparkSession: Using an existing Spark session; only runtime SQL configurations will take effect.
```

```
✅ Data Loaded Successfully. Total Rows: 4988
Schema:
root
 |-- crim: double (nullable = true)
 |-- zn: double (nullable = true)
 |-- indus: double (nullable = true)
 |-- chas: integer (nullable = true)
 |-- nox: double (nullable = true)
 |-- rm: double (nullable = true)
 |-- age: double (nullable = true)
 |-- dis: double (nullable = true)
 |-- rad: integer (nullable = true)
 |-- tax: integer (nullable = true)
 |-- ptratio: double (nullable = true)
 |-- b: double (nullable = true)
 |-- lstat: double (nullable = true)
 |-- medv: double (nullable = true)
```

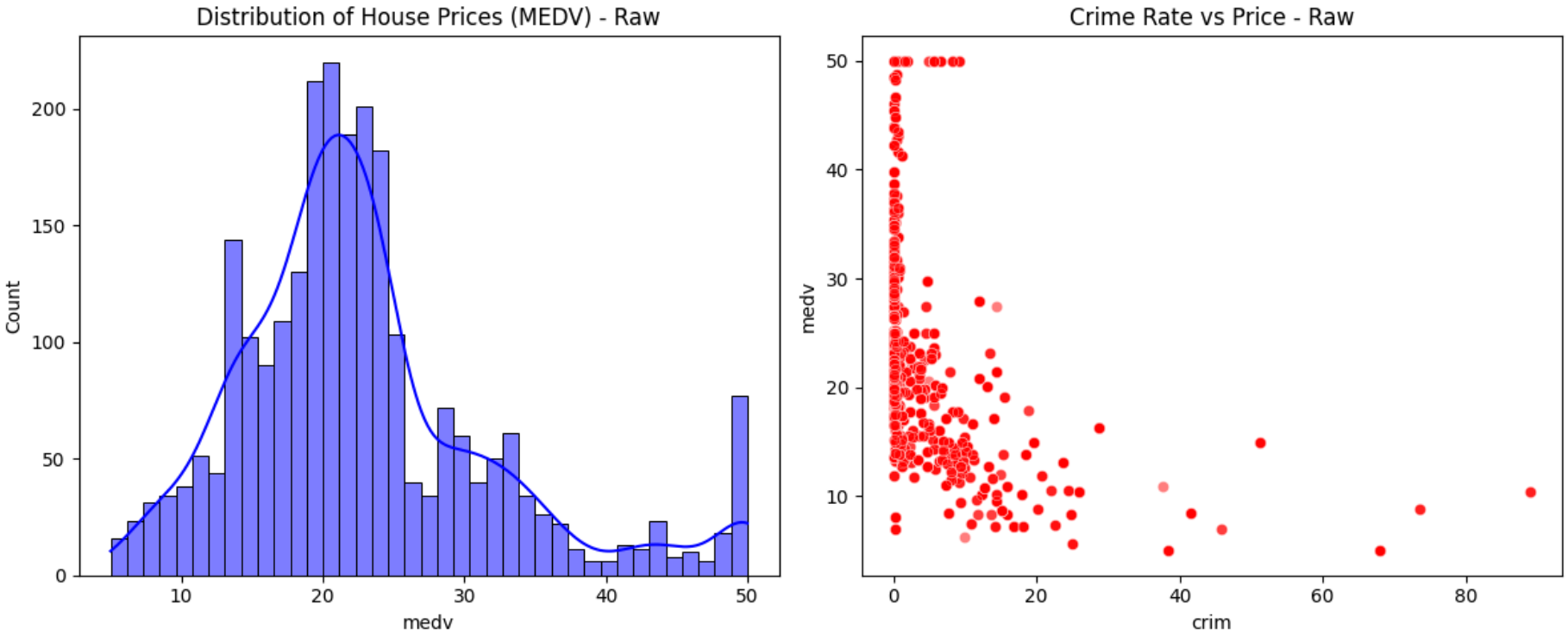
Cell 2: Data Visualization (Before Cleaning)

In [29]:

```python
print("--- Visualizing Raw Data Distribution ---")

# Convert to Pandas for plotting (Sampling 1000 rows to be fast)
pdf = df.sample(False, 0.5, seed=42).toPandas()

plt.figure(figsize=(12, 5))

# Plot 1: Price Distribution
plt.subplot(1, 2, 1)
sns.histplot(pdf['medv'], kde=True, color='blue')
plt.title('Distribution of House Prices (MEDV) - Raw')

# Plot 2: Crime vs Price
plt.subplot(1, 2, 2)
sns.scatterplot(x=pdf['crim'], y=pdf['medv'], color='red', alpha=0.5)
plt.title('Crime Rate vs Price - Raw')

plt.tight_layout()
```

```
plt.show()
print("✅ Visualization Complete. Notice the outliers in Crime Rate!")
```

```
--- Visualizing Raw Data Distribution ---
```




```
✅ Visualization Complete. Notice the outliers in Crime Rate!
```

Cell 3: Outlier Removal

In [30]:
```
print("\n--- Starting Outlier Removal (Iterative 3xIQR) ---")
columns_to_check = ['crim', 'zn', 'chas', 'rm', 'dis', 'ptratio', 'b', 'lstat', 'medv']
print(f"Columns to clean: {columns_to_check}")

df_clean = df

for column in columns_to_check:
    # Calculate Exact Quantiles
    quantiles = df_clean.approxQuantile(column, [0.05, 0.95], 0.0)
    Q1 = quantiles[0]
```

```
    Q3 = quantiles[1]
    IQR = Q3 - Q1

    lower_bound = Q1 - 3 * IQR
    upper_bound = Q3 + 3 * IQR

    # Filter
    prev_count = df_clean.count()
    df_clean = df_clean.filter((col(column) >= lower_bound) & (col(column) <= upper_bound))
    new_count = df_clean.count()

    if prev_count != new_count:
        print(f"   -> Removed {prev_count - new_count} outliers from column: {column}")

print(f"\n✅ Cleaning Complete.")
print(f"Original Rows: {row_count}")
print(f"Final Rows:    {df_clean.count()}")
```

```
--- Starting Outlier Removal (Iterative 3xIQR) ---
Columns to clean: ['crim', 'zn', 'chas', 'rm', 'dis', 'ptratio', 'b', 'lstat', 'medv']
   -> Removed 24 outliers from column: crim
   -> Removed 56 outliers from column: rm

✅ Cleaning Complete.
Original Rows: 4988
Final Rows:    4908
```

Cell 4: Scaling & Feature Engineering

In [31]:

```
print("\n--- Building Feature Engineering Pipeline ---")

# 1. Identify Input Features
feature_cols = df_clean.columns
if 'medv' in feature_cols: feature_cols.remove('medv')
print(f"Input Features: {feature_cols}")

# 2. Define Stages
assembler = VectorAssembler(inputCols=feature_cols, outputCol="raw_features")
scaler = MinMaxScaler(inputCol="raw_features", outputCol="features")

# 3. Fit & Transform
```

```python
print("Fitting Scaler on the WHOLE dataset...")
pipeline = Pipeline(stages=[assembler, scaler])
pipeline_model = pipeline.fit(df_clean)
final_data = pipeline_model.transform(df_clean)

print("✅ Data Scaled (0-1). Preview of Features:")
final_data.select("features", "medv").show(5, truncate=False)
```

```
--- Building Feature Engineering Pipeline ---
Input Features: ['crim', 'zn', 'indus', 'chas', 'nox', 'rm', 'age', 'dis', 'rad', 'tax', 'ptratio', 'b', 'lstat']
Fitting Scaler on the WHOLE dataset...
✅ Data Scaled (0-1). Preview of Features:
+-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------+----+
|features                                                                                                                                                                           |medv|
+-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------+----+
|[0.0,0.18,0.06781524926686218,0.0,0.3148148148148149,0.5775052692086607,0.6416065911431514,0.2692031390664642,0.0,0.2080152671755725,0.2872340425531916,1.0,0.08967991169977926]                |24.0|
|[4.105263734346604E-4,0.0,0.2423020527859238,0.0,0.17283950617283944,0.5479977007089482,0.7826982492276005,0.3489619801944185,0.043478260869565216,0.1049618320610687,0.5531914893617023,1.0,0.20447019867549668]|21.6|
|[4.1013520966769066E-4,0.0,0.2423020527859238,0.0,0.17283950617283944,0.6943858976815481,0.5993820803295572,0.3489619801944185,0.043478260869565216,0.1049618320610687,0.5531914893617023,0.9897372535175752,0.06346578366445917]|34.7|
|[5.094908064779849E-4,0.0,0.06304985337243403,0.0,0.15020576131687244,0.6585552787890402,0.44181256436663235,0.4485445898389546,0.08695652173913043,0.06679389312977099,0.648936170212766,0.9942760603156993,0.033388520971302425]|33.4|
|[0.001226885155100345,0.0,0.06304985337243403,0.0,0.15020576131687244,0.6871048093504504,0.5283213182286304,0.4485445898389546,0.08695652173913043,0.06679389312977099,0.648936170212766,1.0,0.09933774834437085]|36.2|
+-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------+----+
only showing top 5 rows
```

Cell 5: Train/Test Split

In [33]:
```python
print("\n--- Splitting Data ---")
# Using seed=0 to match Sklearn's random_state=0
train_data, test_data = final_data.randomSplit([0.8, 0.2], seed=0)

train_count = train_data.count()
test_count = test_data.count()
```

```
print(f"Training Set: {train_count} rows")
print(f"Testing Set:  {test_count} rows")
```

```
--- Splitting Data ---
Training Set: 3890 rows
Testing Set:  1018 rows
```

Cell 6: Train Random Forest

In [35]:
```
print("\n--- Training Random Forest Regressor ---")
print("Configuration: Trees=100, MaxDepth=30, MaxBins=128")

rf = RandomForestRegressor(
    featuresCol="features",
    labelCol="medv",
    numTrees=100,
    seed=42,
    maxDepth=30,
    maxBins=128
)

rf_model = rf.fit(train_data)
print("✅ Model Trained Successfully!")
```

```
--- Training Random Forest Regressor ---
Configuration: Trees=100, MaxDepth=30, MaxBins=128
```

```
25/11/29 11:24:38 WARN DAGScheduler: Broadcasting large task binary with size 1578.2 KiB
25/11/29 11:24:38 WARN DAGScheduler: Broadcasting large task binary with size 2.5 MiB
25/11/29 11:24:40 WARN DAGScheduler: Broadcasting large task binary with size 3.7 MiB
25/11/29 11:24:41 WARN DAGScheduler: Broadcasting large task binary with size 5.1 MiB
25/11/29 11:24:42 WARN DAGScheduler: Broadcasting large task binary with size 6.7 MiB
25/11/29 11:24:44 WARN DAGScheduler: Broadcasting large task binary with size 8.1 MiB
25/11/29 11:24:46 WARN DAGScheduler: Broadcasting large task binary with size 9.4 MiB
25/11/29 11:24:47 WARN DAGScheduler: Broadcasting large task binary with size 10.5 MiB
25/11/29 11:24:49 WARN DAGScheduler: Broadcasting large task binary with size 11.3 MiB
25/11/29 11:24:50 WARN DAGScheduler: Broadcasting large task binary with size 12.0 MiB
25/11/29 11:24:52 WARN DAGScheduler: Broadcasting large task binary with size 12.4 MiB
25/11/29 11:24:53 WARN DAGScheduler: Broadcasting large task binary with size 12.5 MiB
25/11/29 11:24:54 WARN DAGScheduler: Broadcasting large task binary with size 11.8 MiB
25/11/29 11:24:55 WARN DAGScheduler: Broadcasting large task binary with size 10.9 MiB
25/11/29 11:24:56 WARN DAGScheduler: Broadcasting large task binary with size 9.4 MiB
25/11/29 11:24:57 WARN DAGScheduler: Broadcasting large task binary with size 6.6 MiB
25/11/29 11:24:57 WARN DAGScheduler: Broadcasting large task binary with size 4.4 MiB
25/11/29 11:24:57 WARN DAGScheduler: Broadcasting large task binary with size 1862.6 KiB
```

✅ Model Trained Successfully!

Cell 7: Evaluation & Performance Metrics

In [36]:

```python
print("\n--- Evaluating Model Performance ---")

predictions = rf_model.transform(test_data)

evaluator_r2 = RegressionEvaluator(labelCol="medv", predictionCol="prediction", metricName="r2")
evaluator_rmse = RegressionEvaluator(labelCol="medv", predictionCol="prediction", metricName="rmse")
evaluator_mae = RegressionEvaluator(labelCol="medv", predictionCol="prediction", metricName="mae")

r2 = evaluator_r2.evaluate(predictions)
rmse = evaluator_rmse.evaluate(predictions)
mae = evaluator_mae.evaluate(predictions)
mse = rmse ** 2

print("-" * 30)
print(f"Mean Squared Error (MSE):       {mse:.5f}")
print(f"R-squared (R2):                 {r2:.5f}")
print(f"Mean Absolute Error (MAE):      {mae:.5f}")
```

```
print(f"Root Mean Squared Error (RMSE): {rmse:.5f}")
print("-" * 30)
```

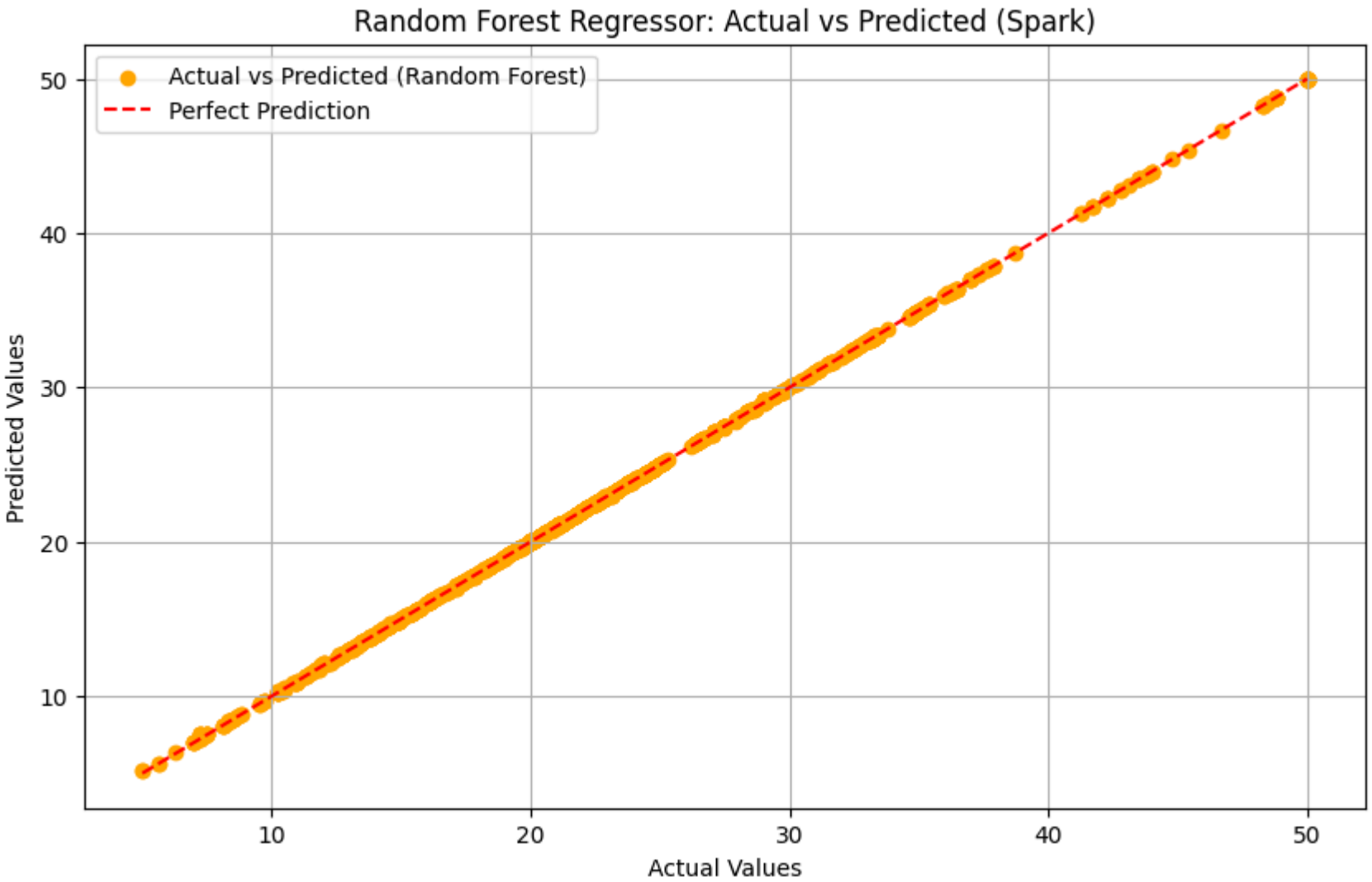
```
--- Evaluating Model Performance ---
------------------------------
Mean Squared Error (MSE):       0.00148
R-squared (R2):                 0.99998
Mean Absolute Error (MAE):      0.01622
Root Mean Squared Error (RMSE): 0.03846
------------------------------
```

Cell 8: Visualization (Actual vs Predicted)

In [37]:

```
print("\n--- Visualizing Predictions ---")

# Convert predictions to Pandas for plotting
pred_pdf = predictions.select("medv", "prediction").toPandas()

plt.figure(figsize=(10, 6))

# Scatter Plot (Orange)
plt.scatter(pred_pdf['medv'], pred_pdf['prediction'], color='orange', label='Actual vs Predicted (Random Forest)')

# Perfect Prediction Line (Red Dashed)
min_val = min(pred_pdf['medv'].min(), pred_pdf['prediction'].min())
max_val = max(pred_pdf['medv'].max(), pred_pdf['prediction'].max())
plt.plot([min_val, max_val], [min_val, max_val], linestyle='--', color='red', label='Perfect Prediction')

plt.title('Random Forest Regressor: Actual vs Predicted (Spark)')
plt.xlabel('Actual Values')
plt.ylabel('Predicted Values')
plt.legend()
plt.grid(True)
plt.show()
print("✅ Plot Generated.")
```

```
--- Visualizing Predictions ---
```



✅ `Plot Generated.`

Cell 9: Save Model & Results

In [39]:
```
output_path = "/home/praveen/BDA/Python_project/boston_output"
print(f"\n--- Saving Results to {output_path} ---")

# Save Model
```

```
model_path = f"{output_path}/rf_model_high_acc"
rf_model.write().overwrite().save(model_path)
print(f"✅ Model saved to: {model_path}")

# Save CSV
csv_path = f"{output_path}/predictions_high_acc_csv"
predictions.select(col("features").cast("string"), col("medv"), col("prediction")) \
    .write.csv(csv_path, header=True, mode="overwrite")
print(f"✅ Predictions saved to: {csv_path}")
```

```
--- Saving Results to /home/praveen/BDA/Python_project/boston_output ---
✅ Model saved to: /home/praveen/BDA/Python_project/boston_output/rf_model_high_acc
✅ Predictions saved to: /home/praveen/BDA/Python_project/boston_output/predictions_high_acc_csv
```

Cell 10: Single Data Point Prediction

In [40]:

```
from pyspark.sql.types import DoubleType

print("--- Predict House Price (Interactive) ---")

# 1. Take User Inputs
# (I cast them to float/int immediately to ensure Spark accepts them)
try:
    crim = float(input("Enter Crime rate in zone (crim): ") or 0.00632)
    zn = float(input("Enter Residential land zone (zn): ") or 18.0)
    indus = float(input("Enter Non-retail business acres (indus): ") or 2.31)
    chas = float(input("Charles River (1=Yes, 0=No) (chas): ") or 0)
    nox = float(input("Nitric oxides concentration (nox): ") or 0.538)
    rm = float(input("Average rooms per dwelling (rm): ") or 6.575)
    age = float(input("Age built prior to 1940 (age): ") or 65.2)
    dis = float(input("Distance to employment centers (dis): ") or 4.09)
    rad = float(input("Accessibility to highways (rad): ") or 1)
    tax = float(input("Property tax rate (tax): ") or 296)
    ptratio = float(input("Pupil-teacher ratio (ptratio): ") or 15.3)
    b = float(input("Black resident proportion (b): ") or 396.9)
    lstat = float(input("Lower status population % (lstat): ") or 4.98)

    # Note: If you just press ENTER for all, it uses the default values (Boston Row #1)

    # 2. Create a Spark DataFrame for this single row
    # We must match the column order used in training
```

```python
    input_data = [(crim, zn, indus, chas, nox, rm, age, dis, rad, tax, ptratio, b, lstat)]
    input_cols = ['crim', 'zn', 'indus', 'chas', 'nox', 'rm', 'age', 'dis', 'rad', 'tax', 'ptratio', 'b', 'lstat']

    new_house_df = spark.createDataFrame(input_data, input_cols)

    # 3. Transform (Scale) the data
    # We use the pipeline_model from the previous cell because it remembers the Min/Max of the original data!
    print("\nScaling input data...")
    processed_house = pipeline_model.transform(new_house_df)

    # 4. Predict
    prediction_result = rf_model.transform(processed_house)

    # 5. Extract the value
    predicted_price = prediction_result.select("prediction").collect()[0][0]

    print("-" * 30)
    print(f"✅ Prediction Value : [{predicted_price:.3f}]")
    print("-" * 30)

except Exception as e:
    print(f"❌ Error during prediction: {e}")
```

```
--- Predict House Price (Interactive) ---
Scaling input data...
[Stage 475:>                                                        (0 + 12) / 12]
------------------------------
✅ Prediction Value : [24.000]
------------------------------
```

In [ ]: